

Experiment 8

AIM: Use pytest to test a Python application.

```
C:\>pip install pytest
Collecting pytest
  Downloading pytest-9.0.2-py3-none-any.whl.metadata (7.6 kB)
Collecting colorama>=0.4 (from pytest)
  Downloading colorama-0.4.6-py2.py3-none-any.whl.metadata (17 kB)
Collecting iniconfig>=1.0.1 (from pytest)
  Downloading iniconfig-2.3.0-py3-none-any.whl.metadata (2.5 kB)
Collecting packaging>=22 (from pytest)
  Downloading packaging-26.0-py3-none-any.whl.metadata (3.3 kB)
Collecting pluggy<2,>=1.5 (from pytest)
  Downloading pluggy-1.6.0-py3-none-any.whl.metadata (4.8 kB)
Collecting pygments>=2.7.2 (from pytest)
  Downloading pygments-2.20.0-py3-none-any.whl.metadata (2.5 kB)
Downloading pytest-9.0.2-py3-none-any.whl (374 kB)
- ----- 374.8/374.8 kB 897.9 kB/s eta 0:00:00
Downloading colorama-0.4.6-py2.py3-none-any.whl (25 kB)
Downloading iniconfig-2.3.0-py3-none-any.whl (7.5 kB)
Downloading packaging-26.0-py3-none-any.whl (74 kB)
----- 74.4/74.4 kB 822.1 kB/s eta 0:00:00
Downloading pluggy-1.6.0-py3-none-any.whl (20 kB)
Downloading pygments-2.20.0-py3-none-any.whl (1.2 MB)
----- 1.2/1.2 MB 758.8 kB/s eta 0:00:00
Installing collected packages: pygments, pluggy, packaging, iniconfig, colorama, pytest
Successfully installed colorama-0.4.6 iniconfig-2.3.0 packaging-26.0 pluggy-1.6.0 pygments-2.20.0 pytest-9.0.2
```

```
[notice] A new release of pip is available: 24.0 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

```
C:\>pytest --version
pytest 9.0.2
```

```
C:\>mkdir Pytest_demo
```

```
C:\>cd Pytest_demo
```

```
C:\Pytest_demo>notepad
ad pythondemo.py
```

```
#pythondemo.py
def add(a, b):
    return a + b
def subtract(a, b):
    return a - b
def multiply(a, b):
    return a * b
def divide(a, b):
    if b == 0:
        raise ValueError("Cannot divide by zero")
    return a / b
def is_even(num):
    return num % 2 == 0
```

```
C:\Pytest_demo>notepad pythondemo.py
# test_pythondemo.py
```

```
import pythondemo
# flassing test
def test_add():
    assert pythondemo.add(2, 3) == 5
# Failing test (Written to fail on purpose as per the manual)
def test_multiply_fail():
    assert pythondemo.multiply(4, 3) == 11
# Exception test
def test_divide_by_zero():
    with pytest.raises(ValueError):
        pythondemo.divide(10, 0)
# Logical test
def test_is_even():
    assert pythondemo.is_even(4) is True
    assert pythondemo.is_even(5) is False
# flarametrized test
@pytest.mark.parametrize("a,b,result", [
    (1, 2, 3),
    (2, 2, 4),
    (5, 5, 11) # failing case
])
def test_add_param(a, b, result):
    assert pythondemo.add(a, b) == result
```

C:\Pytest_demo>pytest -v

```
C:\Pytest_demo>pytest -v
===== test session starts =====
platform win32 -- Python 3.11.9, pytest-9.0.2, pluggy-1.6.0 -- C:\Users\student\AppData\Local\Programs\Python\Python311\python.exe
cachedir: .pytest_cache
rootdir: C:\Pytest_demo
collected 7 items

test_pythondemo.py::test_add PASSED [ 14%]
test_pythondemo.py::test_divide_by_zero PASSED [ 28%]
test_pythondemo.py::test_factorial PASSED [ 42%]
test_pythondemo.py::test_add_param[1-2-3] FAILED [ 57%]
test_pythondemo.py::test_add_param[2-2-4] FAILED [ 71%]
test_pythondemo.py::test_add_param[10-5-15] FAILED [ 85%]
test_pythondemo.py::test_multiply_fail FAILED [100%]

===== FAILURES =====
test_add_param[1-2-3]
test_add_param[2-2-4]
test_add_param[10-5-15]
test_multiply_fail
```

Task 1:

Add a function:

```
def factorial(n):
def factorial(n):
    if n < 0:
        raise ValueError("Factorial is not defined for negative numbers")
    if n == 0 or n == 1:
        return 1
    result = 1
    for i in range(2, n + 1):
        result *= i
    return result
```

Write test cases:

```
import pytest
import pythondemo
# Task 1: Normal test case
def test_factorial_normal():
    assert pythondemo.factorial(5) == 120
# Task 2: Test case for negative numbers (Exception handling)
def test_factorial_negative():
    with pytest.raises(ValueError):
        pythondemo.factorial(-5)
# Task 2: Test case for large inputs
def test_factorial_large_input():
    assert pythondemo.factorial(10) == 3628800
```

```
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test_pythondemo.py::test_add_param[10-5-15] PASSED [ 85%]
test_pythondemo.py::test_multiply_fail FAILED [100%]

===== FAILURES =====
test_multiply_fail
```